

Design and Development of a Low-Cost 2-DOF Haptic Training Interface for Robotic Manipulator Simulation

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1 Background and Motivation

Robotic manipulators are increasingly used in manufacturing, surgical, and remote handling applications. However, operator training often relies on visual feedback alone, limiting spatial understanding and increasing the risk of errors when transitioning to physical systems.

Haptic feedback provides tactile and force based sensations that improve the realism and safety of training by allowing users to feel resistance and contact forces. Commercial haptic devices are often expensive and closed source, which makes them unsuitable for flexible training or academic development.

This project aims to create a low cost modular 2 DOF haptic input device that can send user motion commands to a simulated robotic manipulator and receive realistic force feedback. The system will be designed to support future integration with real robotic hardware, providing an accessible and scalable training platform.

2 Aim

To design, prototype and evaluate a compact low cost two degree of freedom haptic interface that can communicate in real time with a robotic manipulator simulation and provide accurate position sensing and responsive force feedback to the user.

3 Objectives

1. Literature and Technology Review

A review will be carried out to identify existing haptic devices and relevant control methods. This will guide the selection of motors, encoders, and actuation strategies suitable for a low cost modular interface.

2. Physics Simulation Development

A physics based simulation environment will be designed to model manipulator dynamics, haptic interactions, and constraint behavior.

3. Hardware Design

The mechanical and electronic design of the system will be developed, including the selection of hardware components. CAD modeling and circuit design will ensure precise alignment and modularity.

4. Hardware Assembly

The 2 DOF haptic interface will be assembled and wired, focusing on mechanical rigidity, smooth motion, and safe electrical operation. Initial tests will verify the correct motor and encoder functionality.

5. Firmware Development

Firmware will be implemented on the microcontroller to provide Field Oriented Control, encoder polling, and serial communication. The aim will be to achieve stable closed loop torque control with low latency.

6. Firmware–Simulation Integration

The physical prototype will be connected to the existing simulation through a bidirectional serial protocol. This integration will enable real time position tracking and force feedback between the device and the simulated manipulator.

7. Testing and Evaluation

The complete system will be tested to evaluate latency, position accuracy, and control stability. Quantitative results will be compared to simulation outputs to assess fidelity and responsiveness.

8. Documentation and Report

A final project report and supporting documentation will be prepared, summarising the design process, system performance, and key findings.